East West University

Department of Economics

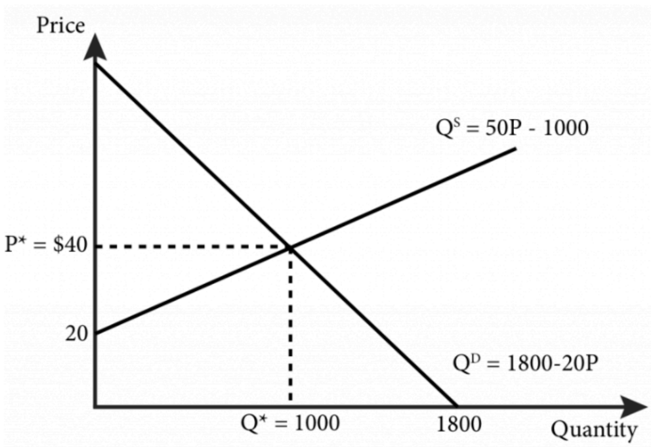
Eco 101 : Principles of Microeconomics

Section 10 & 21

Assignment 02

Fall 2025

1.



Initially, the market for headphones was in equilibrium where the equilibrium price was **$40** and the equilibrium quantity was **1000 units**.

(a) From the diagram, graphically identify the **consumer surplus (CS)** and **producer surplu (PS)** before the imposition of the price ceiling. **[2 marks]**

(b) Then, a **price ceiling of $30** was imposed. Explain how this price ceiling affects producer surplus and consumer surplus. Graphically identify the **new CS**, **new PS**, and the **deadweight loss** created by the price ceiling. **[3 marks]**

2. Suppose the demand function is P=200-2Q and supply function is P=50+Q. If the government introduces a price floor of P=120, find-

(a) Equilibrium price and quantity. **[2 marks]**
(b) Graphically identify and calculate the Producer Surplus and Consumer Surplus before the imposition of price floor. **[3 marks]**
(c) Graphically identify and calculate the Producer Surplus and Consumer Surplus after the imposition of price floor. **[3 marks]**
(d) Graphically identify and calculate the Deadweight loss from the price floor. **[2 marks]**

3. An individual has an income of $500 to spend on two leisure activities: movies and swimming.

- The price of one movie ticket is $50.
- The price of one swimming session is $25.

The individual wants to decide how many movies and swimming sessions to buy with the $500 budget.

Table: Possible Combinations

Option	Movies (at $50 each)	Swimming Sessions (at $25 each)	Total Cost ($)
A	0	20	500
B	2	16	500
C	4	12	500
D	6	8	500
E	8	4	500
F	10	0	500

(a) Draw the individual's budget line**. [3 marks]**

4. Say there is a rational person with a budget of $10. He/she wants to choose a combination of burgers and juice that gives them the maximum possible utility..

Juice		**Burgers**	
Glasses ($2 per glass)	Total Utility (Utils)	No. of burgers ($1 per burger)	Total Utility (Utils)
1	30	1	13
2	55	2	25
3	75	3	36
4	90	4	46

(a) From the given information, find out how many burgers and juices will the consumer choose to maximize his utility(Optimal bundle) **[3 marks]**

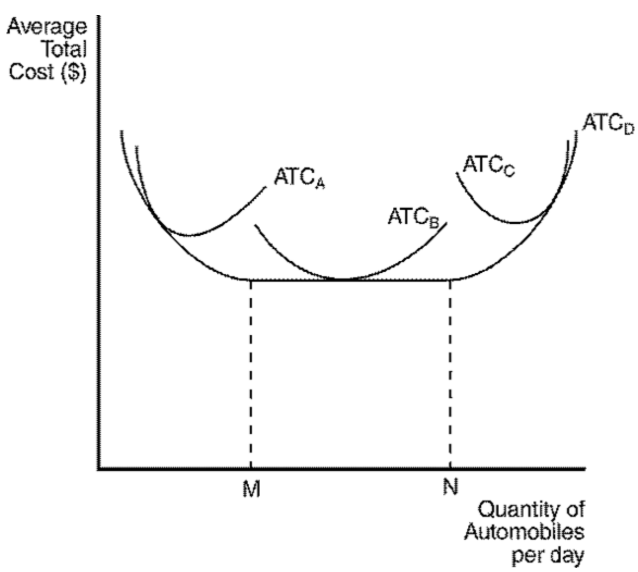
5. The Noodle King Restaurant has daily production costs where each worker earns Tk. 80 per day, and the rent and equipment cost Tk. 250 per day (fixed cost).

Q	MP	L	FC	VC	TC	AFC	AVC	ATC	MC
0	N/A	0				N/A	N/A	N/A	N/A
8		1							
20		2							
36		3							
50		4							
60		5							
64		6							

Where:Q= Quantity of Noodles produced, **MP** = Marginal Product of workers, **L** = Number of workers, **FC** = Fixed Cost, **VC** = Variable Cost, **TC** = Total Cost, **AFC** = Average Fixed Cost, **AVC** = Average Variable Cost, **ATC** = Average Total Cost, **MC** = Marginal Cost

(a) Using the given information, complete the table. **[6 marks]**
(b) Draw a diagram to plot the AFC, AVC, ATC, MC curves using the information provided. **[4 marks]**

6. The figure below depicts average total cost functions for a firm that produces automobiles.



(a) Identify the curve as per the below questions:[3 marks]

i) The short-run ATC of the smallest factory?

ii) The long-run ATC?

iii) The curves the firm can operate on in the long run?

(b) Identify the output ranges where the firm experiences: [3 marks]

i) Economies of scale

ii) Constant returns to scale

iii) Diseconomies of scale

7. The table below shows partial cost data for Speedy Skateboard Co. Some values are missing.

Quantity of Skateboards	Variable Costs (VC)	Total Costs (TC)	Fixed Costs (FC)
0			$50
1	$15		
2		$80	
3	$40		
4		$120	
5	$75		
6		$150	$50

(a) **Fill in all missing values** in the table. **[3 marks]**